FH Innovations Inc. (FHI)

PHASE C

An Enterprise Architecture Proposal

By:

Aquino, Raymond Joaquin

Garrido, Jose Joaquin

Solano, Drew Marco

University of Asia & The Pacific

The Open Group Architecture Framework (TOGAF), 10th Edition

Phase C — Information Systems Architecture

Data Architecture

To resolve current data silos, FHI must centralize master data across five core entities:

Entity	Purpose	Key attributes
Inventory Master	Stock of treatment solutions and supplies across storage and consignments	Item ID, description, unit, supplier, location (storage or dealer), quantity on hand, safety stock, lead time
Dealer & Partner Master	Dealer and supplier profiles	Partner ID, name, site (Location), contacts, consignment and payment terms (Contract), bank/payment notes
Job Order (JO) Ledger	Pre-numbered service transactions	JO number, date, dealer, vehicle Make/Model/Plate No., service type, items used, status (Completed, Pending Further Repairs, Cancelled), cancellation reason, assigned staff
Consignment & Billing Database	Items delivered vs. used vs. billed	PO number, Delivery Receipt number, dealer, item, qty delivered, qty used (from JOs), qty remaining, statement number, amount
Accounts Receivable (AR) Ledger	Invoices, collections and bank deposits	Invoice number, dealer, issue date, terms, due date, aging bracket (0-30, 31-60, 61+), amount, payments, bank account, deposit date, follow-up log

Conceptual data relationships

From	Relationship	To
Dealer & Partner Master	has many	Job Orders
Dealer & Partner Master	receives many	Deliveries (Consignment records)
Inventory Master	is consumed by	Job Order line items
Job Order	posts usage to	Consignment & Billing record
Consignment & Billing record	generates	Invoice in AR Ledger
AR Ledger invoice	has many	Payments / bank deposits

Application Architecture

Given FHI's status as an MSME with 27 employees and no existing IT architecture, a lightweight, cloud-based or local-server-hosted ERP or Inventory & Invoicing SaaS is recommended, comprising three modules:

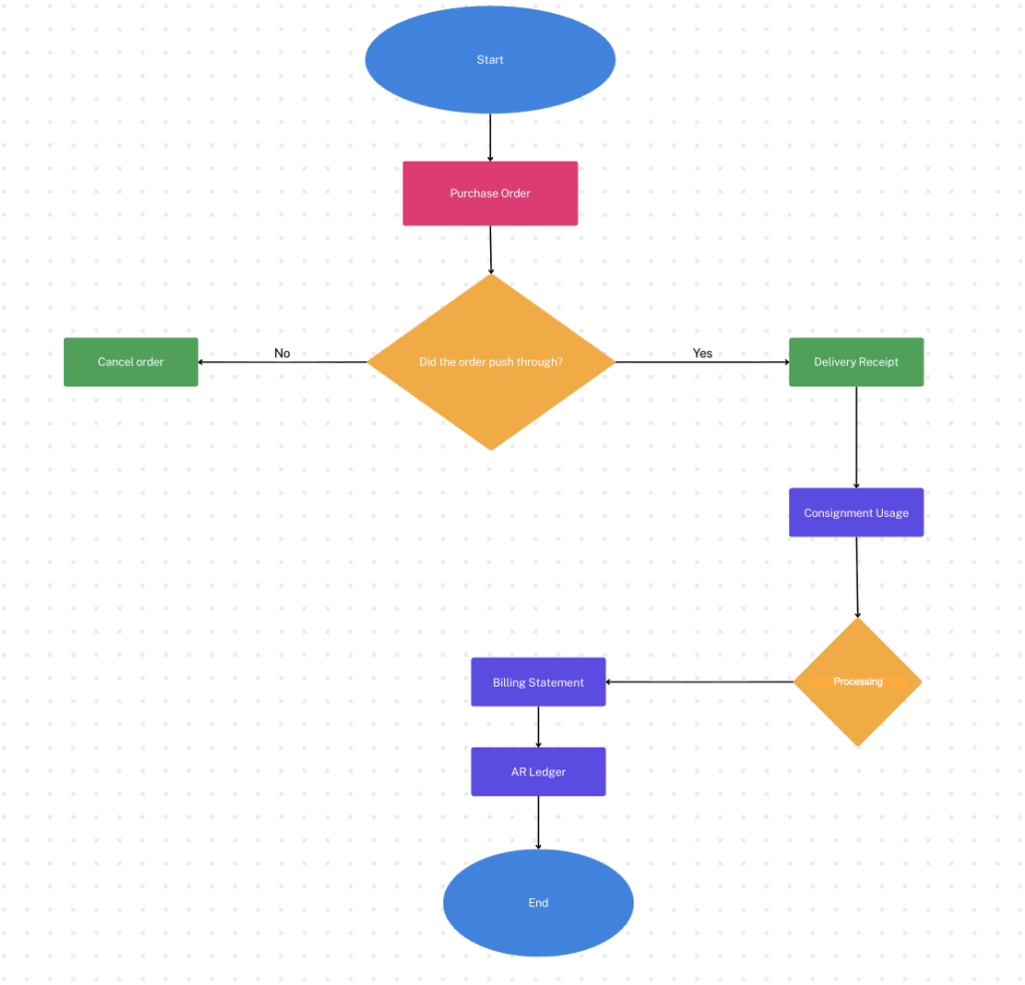
Application	Function	Data entities used
Inventory Management Module	Real-time stock, PO, delivery receipt tracking, reorder alerts	Inventory Master, Dealer & Partner Master, Consignment & Billing
Service & Job Order Module	Mobile/tablet JO logging (offline-first)	JO Ledger, Inventory Master
Billing & AR Module	Statements, aging reports, collection calendar, bank reconciliation	Consignment & Billing, AR Ledger
Backup Service	Daily automated backup	All

Interface catalog

Interface	From	To	Content	Mode
I1	Inventory module	Service & JO module	Stock and delivery events	Internal
I2	Service & JO module	Billing & AR module	Verified job completion and usage	Internal
I3	Billing & AR module	Dealers	Consignment statements	Document/email handoff
I4	Bank accounts	Billing & AR module	Deposit records	Manual upload or statement import
I5	Inventory module	Suppliers	Purchase Orders	Email or messaging channel

Data Entity / Data Flow

Data flows in a single directional chain that mirrors the Three-Way Consignment Reconciliation principle:



Application Communication Diagram

The Inventory Management Module publishes stock and delivery events consumed by the Service & Job Order Module, which in turn confirms completed services to the Billing & AR Module. The Billing & AR Module reconciles against multi-bank deposit feeds to close the loop from job completion to cash collection.

Role / Application matrix

Role	Inventory	Service & JO	Billing & AR
Administrator	Full	Validate	Full
Service Technician	View stock	Create and update JOs	None
President	Dashboard	Dashboard	Dashboard
Dealer Service Advisor	None	Request only (external)	Receives statements

Information Use-Case Matrix

Data Object	Primary Business Role(s)	Use Case
Purchase Orders (PO)	Administrator, President	Trigger and track supplier procurement.
Job Orders (JO)	Service Technician, Administrator	Log and verify service execution per vehicle.
Consignment Ledger	Administrator	Reconcile items delivered vs. items sold per dealer.
AR Aging Buckets	Administrator, President	Drive collection follow-up and cash-flow visibility.